

PROBLEMS Week: 23 April

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

FROM CHAPTER 13: For week of 23 April 2018

- P13.30** (a) $-4.77 \times 10^9 \text{ J}$
(b), (c) The satellite and Earth exert forces of equal magnitude on each other, directed downward on the satellite and upward on Earth.
The magnitude of this force is 569 N

- P13.33** (a) $1.84 \times 10^9 \text{ kg/m}^3$
(b) $3.27 \times 10^6 \text{ m/s}^2$
(c) $-2.08 \times 10^{13} \text{ J}$

- P13.35** (a) $-1.67 \times 10^{-14} \text{ J}$

Each particle feels a net force of attraction toward the midpoint between the other two. Each moves toward the center of the
(b) triangle with the same acceleration. They collide simultaneously at the center of the triangle.

FROM CHAPTER 8: For week of 23 April 2018

- P8.5 (a) $v_A = \sqrt{3.00gR}$ (b) 0.098N downward
P8.7 (a) 4.43 m/s (b) 5.0 m
P8.15 (a) 0.791 m/s (b) 0.531 m/s
P8.22 3.74 m/s
P8.23 (a) $\Delta K = -160 \text{ J}$ (b) $\Delta U = 73.5 \text{ J}$ (c) $f_k = 28.8 \text{ N}$ (d) $\mu_k =$
P8.26 The air resistance acts like a frictional force (a) 24.5 m/s (b) E:
tion (convert your answer in (a)) (c) 206 m (d) Explanation

See Section 6.4, Model 2 on page 164

- (d) The air drag is proportional to the square of the skydiver's speed, so it will change quite a bit. It will be larger than her 784-N weight only after the chute is opened. It will be nearly equal to 784 N before she opens the chute and again before she touches down whenever she moves near terminal speed.

P8.31 $4.5 \times 10^3 \text{ N}$

P8.45 $H = h + \frac{d^2}{4h}$

P8.47 (a) $K = 2 + 24t^2 + 72t^4$ (b) $a = 12t, \sum F = 48t$ (c) $P = 48t + 288t^3$

(d) 1250 J

P8.63 $\mu_k = 0.328$

P8.64 $v = 1.24 \text{ m/s}$

P8.73 Proof that $T_{\text{bottom}} = T_{\text{top}} + 6mg$